

Do Carmo §1.3 Problem Set

Exercise 1. Show that the tangent lines to the regular parametrized curve $\alpha(t) = (3t, 3t^2, 2t^3)$ make a constant angle with the line $y = 0, z = x$.

Solution. First, we compute the tangent vector $\alpha'(t)$ by differentiating each component with respect to t :

$$\alpha'(t) = (3, 6t, 6t^2)$$

Next, we find the magnitude (length) of the tangent vector:

$$|\alpha'(t)| = \sqrt{3^2 + (6t)^2 + (6t^2)^2} = \sqrt{9 + 36t^2 + 36t^4}$$

We can factor out 9 and rewrite the expression under the square root as a perfect square:

$$|\alpha'(t)| = \sqrt{9(1 + 4t^2 + 4t^4)} = 3\sqrt{(1 + 2t^2)^2} = 3(1 + 2t^2)$$

A direction vector for the line described by $y = 0, z = x$ is given by $\mathbf{u} = (1, 0, 1)$. The magnitude of this direction vector is:

$$|\mathbf{u}| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

The cosine of the angle θ between the tangent vector $\alpha'(t)$ and the line vector $\mathbf{u}$ is determined by their dot product formula:

$$\cos \theta = \frac{\alpha'(t) \cdot \mathbf{u}}{|\alpha'(t)| |\mathbf{u}|}$$

Computing the dot product in the numerator:

$$\alpha'(t) \cdot \mathbf{u} = (3)(1) + (6t)(0) + (6t^2)(1) = 3 + 6t^2 = 3(1 + 2t^2)$$

Substituting these values back into the angle formula:

$$\cos \theta = \frac{3(1 + 2t^2)}{3(1 + 2t^2)\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Since $\cos \theta = \frac{1}{\sqrt{2}}$, the angle is constantly $\theta = \frac{\pi}{4}$ (or 45°), which is completely independent of t . □

Exercise 2. A circular disk of radius 1 in the plane xy rolls without slipping along the x -axis. The figure described by a point of the circumference of the disk is called a cycloid (Fig. 1-7).

*a. Obtain a parametrized curve $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ the trace of which is the cycloid, and determine its singular points.

b. Compute the arc length of the cycloid corresponding to a complete rotation of the disk.

Solution. a. Let t represent the angle (in radians) through which the disk has rotated. Since the disk rolls without slipping along the x -axis, the center of the disk at time t has translated horizontally by a distance equal to the arc length rolled, positioning it at $(t, 1)$.

Taking the tracing point initially at the origin $(0, 0)$, its position relative to the center of the disk is given by $(-\sin t, -\cos t)$. Adding the center translation coordinates yields the standard parametrization:

$$\alpha(t) = (t - \sin t, 1 - \cos t)$$

To determine the singular points, we find where the velocity vector $\alpha'(t)$ vanishes:

$$\alpha'(t) = (1 - \cos t, \sin t) = (0, 0)$$

This requires both components to equal zero simultaneously:

$$1 - \cos t = 0 \implies \cos t = 1 \quad \text{and} \quad \sin t = 0$$

This system is satisfied if and only if t is an integer multiple of 2π . Thus, the singular points (cusps along the ground) occur at:

$$t = 2k\pi \quad \text{for } k \in \mathbb{Z}$$

- b. A complete rotation corresponds to the parameter interval $t \in [0, 2\pi]$. The arc length s is given by the integral of the speed:

$$s = \int_0^{2\pi} |\alpha'(t)| dt = \int_0^{2\pi} \sqrt{(1 - \cos t)^2 + \sin^2 t} dt$$

Expanding and simplifying the integrand using the Pythagorean identity $\cos^2 t + \sin^2 t = 1$:

$$|\alpha'(t)| = \sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t} = \sqrt{2 - 2\cos t}$$

Using the trigonometric half-angle identity $1 - \cos t = 2\sin^2\left(\frac{t}{2}\right)$:

$$|\alpha'(t)| = \sqrt{4\sin^2\left(\frac{t}{2}\right)} = 2\left|\sin\left(\frac{t}{2}\right)\right|$$

Since $\sin\left(\frac{t}{2}\right) \geq 0$ on the interval $[0, 2\pi]$, we drop the absolute value bars and evaluate the integral:

$$s = \int_0^{2\pi} 2\sin\left(\frac{t}{2}\right) dt = \left[-4\cos\left(\frac{t}{2}\right)\right]_0^{2\pi} = -4\cos(\pi) - (-4\cos 0) = 4 + 4 = 8$$

□

Exercise 3. Let $OA = 2a$ be the diameter of a circle S^1 and OY and AV be the tangents to S^1 at O and A , respectively. A half-line r is drawn from O which meets the circle S^1 at C and the line AV at B . On OB mark off the segment $Op = CB$. If we rotate r about O , the point p will describe a curve called the *cisoid of Diocles*. By taking OA as the x -axis and OY as the y -axis, prove that:

- The trace of $\alpha(t) = \left(\frac{2at^2}{1+t^2}, \frac{2at^3}{1+t^2} \right)$, $t \in \mathbb{R}$, is the *cisoid of Diocles* ($t = \tan \theta$).
- The origin $(0,0)$ is a singular point of the *cisoid*.
- As $t \rightarrow \infty$, $\alpha(t)$ approaches the line $x = 2a$, and $\alpha'(t) \rightarrow (2a,0)$ is false; rather, the curve approaches the vertical line asymptotically.

Solution. a. Placing the circle S^1 with diameter $2a$ along the x -axis means its center is at $(a,0)$, yielding the circle equation $x^2 + y^2 = 2ax$. Let the half-line r making an angle θ with the x -axis be written in terms of its slope parameter $t = \tan \theta$, giving the line equation $y = tx$.

To find the intersection point C with the circle, substitute $y = tx$ into the circle equation:

$$x^2 + (tx)^2 = 2ax \implies x(1+t^2) = 2a \implies x_C = \frac{2a}{1+t^2}, \quad y_C = \frac{2at}{1+t^2}$$

The vertical tangent line AV at $A(2a,0)$ has the equation $x = 2a$. Its intersection point B with the line $y = tx$ is:

$$x_B = 2a, \quad y_B = 2at$$

By definition of the geometric construction, the vector components of the point $p = (x,y)$ satisfy $\vec{Op} = \vec{OB} - \vec{OC}$. Subtracting the coordinate values reveals:

$$\begin{aligned} x &= x_B - x_C = 2a - \frac{2a}{1+t^2} = \frac{2a(1+t^2) - 2a}{1+t^2} = \frac{2at^2}{1+t^2} \\ y &= y_B - y_C = 2at - \frac{2at}{1+t^2} = \frac{2at(1+t^2) - 2at}{1+t^2} = \frac{2at^3}{1+t^2} \end{aligned}$$

This matches the given coordinate map $\alpha(t)$.

- We find the derivative components $\alpha'(t) = (x'(t), y'(t))$ using the quotient rule:

$$\begin{aligned} x'(t) &= \frac{4at(1+t^2) - 2at^2(2t)}{(1+t^2)^2} = \frac{4at}{(1+t^2)^2} \\ y'(t) &= \frac{6at^2(1+t^2) - 2at^3(2t)}{(1+t^2)^2} = \frac{6at^2 + 2at^4}{(1+t^2)^2} \end{aligned}$$

Evaluating the derivative vector at the origin, which corresponds to $t = 0$:

$$\alpha'(0) = \left(\frac{0}{1}, \frac{0}{1} \right) = (0,0)$$

Since the velocity vector completely vanishes at $t = 0$, the origin is a singular point.

c. We take limits as the parameter t approaches infinity:

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} \frac{2at^2}{1+t^2} = 2a$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} \frac{2at^3}{1+t^2} = \pm\infty$$

The horizontal coordinate approaches the bounded threshold value $x = 2a$ while the vertical coordinate tends to infinity, confirming that the vertical line $x = 2a$ acts as a geometric asymptote.

□

Exercise 4. Let $\alpha : (0, \pi) \rightarrow \mathbb{R}^2$ be given by

$$\alpha(t) = \left(\sin t, \cos t + \log \tan \frac{t}{2} \right),$$

where t is the angle that the y -axis makes with the vector $\alpha'(t)$. The trace of α is called the *tractrix* (Fig. 1-9). Show that:

- α is a differentiable parametrized curve, regular except at $t = \pi/2$.
- The length of the segment of the tangent of the tractrix between the point of tangency and the y -axis is constantly equal to 1.

Solution. a. To determine where the curve is regular, we differentiate each component function of $\alpha(t) = (x(t), y(t))$ with respect to t .

The derivative of the x -component is:

$$x'(t) = \frac{d}{dt}(\sin t) = \cos t$$

The derivative of the y -component is found using the chain rule:

$$y'(t) = \frac{d}{dt} \left(\cos t + \log \tan \frac{t}{2} \right) = -\sin t + \frac{1}{\tan \frac{t}{2}} \cdot \sec^2 \left(\frac{t}{2} \right) \cdot \frac{1}{2}$$

Using trigonometric identities to simplify the logarithmic derivative term:

$$\frac{1}{\tan \frac{t}{2}} \cdot \sec^2 \left(\frac{t}{2} \right) \cdot \frac{1}{2} = \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} \cdot \frac{1}{\cos^2 \frac{t}{2}} \cdot \frac{1}{2} = \frac{1}{2 \sin \frac{t}{2} \cos \frac{t}{2}} = \frac{1}{\sin t}$$

Combining the terms for $y'(t)$ yields:

$$y'(t) = -\sin t + \frac{1}{\sin t} = \frac{1 - \sin^2 t}{\sin t} = \frac{\cos^2 t}{\sin t}$$

Thus, the velocity vector is given by:

$$\alpha'(t) = \left(\cos t, \frac{\cos^2 t}{\sin t} \right)$$

A parametrized curve is regular where $\alpha'(t) \neq (0, 0)$. Setting $\alpha'(t) = (0, 0)$ requires:

$$\cos t = 0 \quad \text{and} \quad \frac{\cos^2 t}{\sin t} = 0$$

Within the given open interval $t \in (0, \pi)$, $\cos t = 0$ has exactly one solution: $t = \frac{\pi}{2}$. Evaluating at this point shows $\alpha'(\frac{\pi}{2}) = (0, 0)$. Therefore, the curve is differentiable everywhere on its domain and regular everywhere except at the singular point $t = \frac{\pi}{2}$.

- b. The tangent line at any regular point $\alpha(t)$ is directed along the velocity vector $\alpha'(t) = \left(\cos t, \frac{\cos^2 t}{\sin t}\right)$. We can scale this vector by factoring out $\cos t$ to get a simpler parallel direction vector $\mathbf{u} = (1, \cot t)$.

The point-slope equation of the tangent line passing through the point of tangency $P_t = (x(t), y(t))$ is:

$$Y - y(t) = \cot t \cdot (X - x(t))$$

We find the intersection of this tangent line with the vertical y -axis by setting $X = 0$:

$$Y_{\text{int}} - y(t) = \cot t \cdot (0 - x(t)) = -x(t) \cot t$$

Substituting $x(t) = \sin t$ into this expression:

$$Y_{\text{int}} - y(t) = -\sin t \cdot \frac{\cos t}{\sin t} = -\cos t \implies Y_{\text{int}} = y(t) - \cos t$$

Thus, the intersection point on the y -axis is $P_0 = (0, y(t) - \cos t)$.

Finally, we calculate the geometric distance between the point of tangency $P_t = (x(t), y(t))$ and the y -axis intersection point P_0 :

$$\text{Length} = \sqrt{(x(t) - 0)^2 + (y(t) - [y(t) - \cos t])^2}$$

$$\text{Length} = \sqrt{(x(t))^2 + (\cos t)^2} = \sqrt{\sin^2 t + \cos^2 t} = 1$$

This proves that the length of the tangent segment intercepted between the curve and the y -axis is constantly equal to 1.

□

Exercise 5. Let $\alpha : (-1, +\infty) \rightarrow \mathbb{R}^2$ be given by

$$\alpha(t) = \left(\frac{3at}{1+t^3}, \frac{3at^2}{1+t^3} \right).$$

Prove that:

- For $t = 0$, α is tangent to the x -axis.
- As $t \rightarrow +\infty$, $\alpha(t) \rightarrow (0, 0)$ and $\alpha'(t) \rightarrow (0, 0)$.
- Take the curve with the opposite orientation. Now, as $t \rightarrow -1$, the curve and its tangent approach the line $x + y + a = 0$.

Solution. a. First, we evaluate the velocity vector $\alpha'(t) = (x'(t), y'(t))$ using the quotient rule:

$$x'(t) = \frac{3a(1+t^3) - 3at(3t^2)}{(1+t^3)^2} = \frac{3a(1-2t^3)}{(1+t^3)^2}$$

$$y'(t) = \frac{6at(1+t^3) - 3at^2(3t^2)}{(1+t^3)^2} = \frac{3at(2-t^3)}{(1+t^3)^2}$$

Evaluating the velocity components at $t = 0$:

$$\alpha'(0) = \left(\frac{3a(1)}{1}, \frac{0}{1} \right) = (3a, 0)$$

Since $y'(0) = 0$ and $x'(0) = 3a \neq 0$, the velocity vector points strictly along the horizontal direction, which proves that the curve is tangent to the x -axis at $t = 0$.

b. We take the limit of the position coordinates as $t \rightarrow +\infty$:

$$\lim_{t \rightarrow +\infty} x(t) = \lim_{t \rightarrow +\infty} \frac{3at}{1+t^3} = 0, \quad \lim_{t \rightarrow +\infty} y(t) = \lim_{t \rightarrow +\infty} \frac{3at^2}{1+t^3} = 0$$

Thus, $\alpha(t) \rightarrow (0, 0)$. Next, we take the limit of the velocity components using the derivatives from part (a):

$$\lim_{t \rightarrow +\infty} x'(t) = \lim_{t \rightarrow +\infty} \frac{3a - 6at^3}{1 + 2t^3 + t^6} = 0, \quad \lim_{t \rightarrow +\infty} y'(t) = \lim_{t \rightarrow +\infty} \frac{6at - 3at^4}{1 + 2t^3 + t^6} = 0$$

This confirms that $\alpha'(t) \rightarrow (0, 0)$ as $t \rightarrow +\infty$.

c. To show that the curve approaches the line $x + y + a = 0$ as $t \rightarrow -1$, we examine the sum of the coordinates:

$$x(t) + y(t) = \frac{3at + 3at^2}{1 + t^3} = \frac{3at(1+t)}{(1+t)(1-t+t^2)} = \frac{3at}{1-t+t^2}$$

Taking the limit as $t \rightarrow -1$:

$$\lim_{t \rightarrow -1} (x(t) + y(t)) = \frac{3a(-1)}{1 - (-1) + (-1)^2} = \frac{-3a}{3} = -a$$

Rearranging terms gives $\lim_{t \rightarrow -1} (x(t) + y(t) + a) = 0$, which proves that the curve approaches the line.

To check the behavior of the tangent line, we evaluate the limit of the tangent slope $m(t) = \frac{y'(t)}{x'(t)}$ using the derivatives from part (a):

$$m(t) = \frac{3at(2-t^3)/(1+t^3)^2}{3a(1-2t^3)/(1+t^3)^2} = \frac{t(2-t^3)}{1-2t^3}$$

Taking the limit as $t \rightarrow -1$:

$$\lim_{t \rightarrow -1} m(t) = \frac{-1(2 - (-1))}{1 - 2(-1)} = \frac{-3}{3} = -1$$

Since the line $x + y + a = 0$ has a slope of -1 , the directional slope of the curve's tangent lines perfectly aligns with the slope of the asymptote line as $t \rightarrow -1$.

□

Exercise 6. Let $\alpha(t) = (ae^{bt} \cos t, ae^{bt} \sin t)$, $t \in \mathbb{R}$, a and b constants, $a > 0$, $b < 0$, be a parametrized curve.

- a. Show that as $t \rightarrow +\infty$, $\alpha(t)$ approaches the origin 0, spiraling around it.
- b. Show that $\alpha'(t) \rightarrow (0, 0)$ as $t \rightarrow +\infty$ and that

$$\lim_{t \rightarrow +\infty} \int_{t_0}^t |\alpha'(t)| dt$$

is finite; that is, α has finite arc length in $[t_0, \infty)$.

Solution. a. We can rewrite the curve coordinates in polar form $(r(t), \theta(t))$ where $x = r \cos \theta$ and $y = r \sin \theta$. Matching this with the given parametrization shows:

$$r(t) = ae^{bt}, \quad \theta(t) = t$$

Since $b < 0$, the exponential radius function decreases continuously, yielding the limit:

$$\lim_{t \rightarrow +\infty} r(t) = \lim_{t \rightarrow +\infty} ae^{bt} = 0$$

As $t \rightarrow +\infty$, the radius shrinks to zero while the polar angle $\theta = t$ increases indefinitely, forcing the trajectory to tightly revolve around the origin an infinite number of times. This proves that the curve spirals into the origin.

- b. We differentiate the component functions using the product rule:

$$x'(t) = abe^{bt} \cos t - ae^{bt} \sin t = ae^{bt}(b \cos t - \sin t)$$

$$y'(t) = abe^{bt} \sin t + ae^{bt} \cos t = ae^{bt}(b \sin t + \cos t)$$

Since $b < 0$, $\lim_{t \rightarrow +\infty} ae^{bt} = 0$. The terms inside the parentheses remain strictly bounded between $\pm(|b| + 1)$. By the squeeze theorem, the product of a vanishing term and a bounded term goes to zero:

$$\lim_{t \rightarrow +\infty} x'(t) = 0, \quad \lim_{t \rightarrow +\infty} y'(t) = 0 \implies \lim_{t \rightarrow +\infty} \alpha'(t) = (0, 0)$$

To calculate the total arc length, we compute the magnitude of the velocity vector:

$$|\alpha'(t)| = \sqrt{[ae^{bt}(b \cos t - \sin t)]^2 + [ae^{bt}(b \sin t + \cos t)]^2}$$

$$|\alpha'(t)| = ae^{bt} \sqrt{b^2 \cos^2 t - 2b \sin t \cos t + \sin^2 t + b^2 \sin^2 t + 2b \sin t \cos t + \cos^2 t}$$

Factoring and simplifying via $\cos^2 t + \sin^2 t = 1$:

$$|\alpha'(t)| = ae^{bt} \sqrt{b^2(\cos^2 t + \sin^2 t) + (\sin^2 t + \cos^2 t)} = ae^{bt} \sqrt{b^2 + 1}$$

Now we evaluate the improper arc length integral from t_0 to $+\infty$:

$$\text{Arc Length} = \int_{t_0}^{+\infty} ae^{bt}\sqrt{b^2+1} dt = a\sqrt{b^2+1} \left[\frac{e^{bt}}{b} \right]_{t_0}^{+\infty}$$

Taking the limit at the upper bound where $b < 0$:

$$\text{Arc Length} = a\sqrt{b^2+1} \left(0 - \frac{e^{bt_0}}{b} \right) = -\frac{a\sqrt{b^2+1}}{b} e^{bt_0} = \frac{a\sqrt{b^2+1}}{|b|} e^{bt_0}$$

Since a , b , and t_0 are real constants, this evaluated limit expression is finite, confirming that the infinite spiral path has a finite arc length. □

Exercise 7. 7. A map $\alpha : I \rightarrow \mathbb{R}^3$ is called a curve of class C^k if each of the coordinate functions in the expression $\alpha(t) = (x(t), y(t), z(t))$ has continuous derivatives up to order k . If α is merely continuous, we say that α is of class C^0 . A curve α is called simple if the map α is one-to-one. Thus, the curve in Example 3 of Sec. 1-2 is not simple.

Let $\alpha : I \rightarrow \mathbb{R}^3$ be a simple curve of class C^0 . We say that α has a weak tangent at $t = t_0 \in I$ if the line determined by $\alpha(t_0 + h)$ and $\alpha(t_0)$ has a limit position when $h \rightarrow 0$. We say that α has a strong tangent at $t = t_0$ if the line determined by $\alpha(t_0 + h)$ and $\alpha(t_0 + k)$ has a limit position when $h, k \rightarrow 0$. Show that:

a. $\alpha(t) = (t^3, t^2)$, $t \in \mathbb{R}$, has a weak tangent but not a strong tangent at $t = 0$.

*b. If $\alpha : I \rightarrow \mathbb{R}^3$ is of class C^1 and regular at $t = t_0$, then it has a strong tangent at $t = t_0$.

c. The curve given by

$$\alpha(t) = \begin{cases} (t^2, t^2), & t \geq 0, \\ (t^2, -t^2), & t \leq 0, \end{cases}$$

is of class C^1 but not of class C^2 . Draw a sketch of the curve and its tangent vectors.

Solution. a. A line passing through two points $\alpha(t_1)$ and $\alpha(t_2)$ is directed along the secant vector $\vec{v} = \alpha(t_2) - \alpha(t_1)$.

To check for a weak tangent at $t_0 = 0$, we look at the secant line through $\alpha(0) = (0, 0)$ and $\alpha(h) = (h^3, h^2)$. The secant vector is:

$$\vec{v}_{\text{weak}} = (h^3, h^2)$$

Scaling this vector by $1/h^2$ to look at its direction yields $\vec{u}_{\text{weak}} = (h, 1)$. Taking the limit as $h \rightarrow 0$:

$$\lim_{h \rightarrow 0} \vec{u}_{\text{weak}} = (0, 1)$$

Because this direction limit exists, the curve has a well-defined vertical weak tangent line (the y -axis) at the origin.

To check for a strong tangent line, we evaluate the secant vector between two independent neighboring points $\alpha(h) = (h^3, h^2)$ and $\alpha(k) = (k^3, k^2)$:

$$\vec{v}_{\text{strong}} = \alpha(h) - \alpha(k) = (h^3 - k^3, h^2 - k^2) = (h - k)(h^2 + hk + k^2, h + k)$$

Assuming $h \neq k$, we clear the common factor $(h - k)$ to look at the direction vector:

$$\vec{u}_{\text{strong}} = (h^2 + hk + k^2, h + k)$$

Let us evaluate the limiting behavior of this direction along two different paths toward the origin:

- Along the path $k = -h$: $\vec{u}_{\text{strong}} = (h^2 - h^2 + h^2, 0) = (h^2, 0)$, which scales to the horizontal direction $(1, 0)$.
- Along the path $k = 0$: $\vec{u}_{\text{strong}} = (h^2, h)$, which scales to $(h, 1)$ and yields the vertical direction $(0, 1)$ as $h \rightarrow 0$.

Since the limiting direction depends on the path taken by h and k , the limit position does not exist. Thus, α does not possess a strong tangent at $t = 0$.

- *b. Let α be a regular C^1 curve, meaning $\alpha'(t)$ is continuous and $\alpha'(t_0) \neq \mathbf{0}$. By the Mean Value Theorem applied to each coordinate function, there exist independent intermediate parameters c_1, c_2, c_3 strictly between $(t_0 + k)$ and $(t_0 + h)$ such that:

$$\alpha(t_0 + h) - \alpha(t_0 + k) = (h - k) \cdot (x'(c_1), y'(c_2), z'(c_3))$$

As $h, k \rightarrow 0$, the squeeze theorem forces all intermediate values $c_i \rightarrow t_0$. Because the derivative functions are continuous (C^1), we can evaluate the limit directly:

$$\lim_{h, k \rightarrow 0} (x'(c_1), y'(c_2), z'(c_3)) = (x'(t_0), y'(t_0), z'(t_0)) = \alpha'(t_0)$$

Since the curve is regular, $\alpha'(t_0) \neq \mathbf{0}$, meaning the secant lines converge on a unique limit direction parallel to the non-zero velocity vector $\alpha'(t_0)$. This guarantees the existence of a strong tangent.

- c. We check the differentiability at the interface point $t = 0$:

- **First Derivative:** Differentiating both branches yields:

$$\alpha'(t) = \begin{cases} (2t, 2t), & t > 0 \\ (2t, -2t), & t < 0 \end{cases}$$

Evaluating one-sided limits at zero gives $\lim_{t \rightarrow 0^+} \alpha'(t) = (0, 0)$ and $\lim_{t \rightarrow 0^-} \alpha'(t) = (0, 0)$. Since both matching limits equal the direct derivative $\alpha'(0) = (0, 0)$, the curve is of class C^1 .

- **Second Derivative:** Differentiating a second time yields:

$$\alpha''(t) = \begin{cases} (2, 2), & t > 0 \\ (2, -2), & t < 0 \end{cases}$$

The right-hand limit at zero is $(2, 2)$, while the left-hand limit is $(2, -2)$. Because these one-sided limits do not match, $\alpha''(0)$ does not exist, proving the curve is not of class C^2 .

□

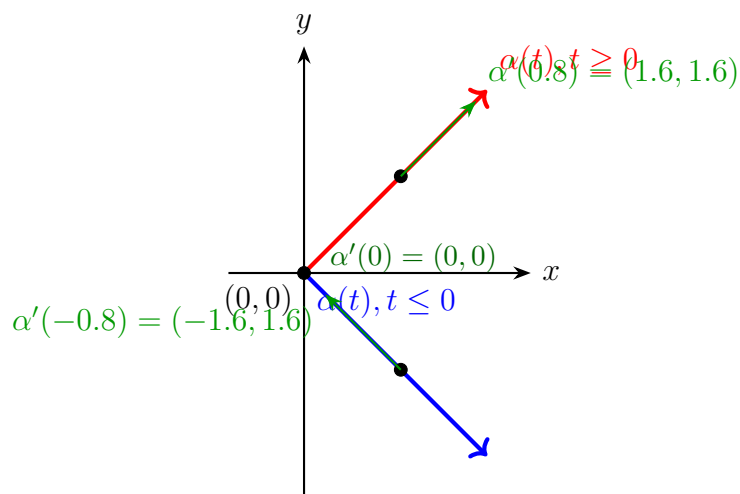


Figure 1: Sketch of the piecewise curve $\alpha(t)$ and its changing tangent vectors. The curve tracks down the ray $y = -x$ in the fourth quadrant, halts at a cusp at the origin, and goes out along $y = x$ in the first quadrant.